

Homework 3

- **Submit your solutions electronically on the course Gradescope site as PDF files.** If you plan to typeset your solutions, please use the $\text{\LaTeX}$ solution template on the course web site. If you must submit scanned handwritten solutions, please use a black pen on blank white paper and a high-quality scanner app (or an actual scanner, not just a phone camera). We will mark difficult to read solutions as incorrect and move on.
- **Every homework problem must be done *individually*.** Each problem needs to be submitted to Gradescope before midnight of the due date which can be found on the course website: <https://ecealgo.com/su26/homeworks.html>.
- For nearly every problem, **we have covered all the requisite knowledge required to complete a homework assignment prior to the “assigned” date.** This means that there is no reason not to begin a homework assignment as soon as it is assigned. Starting a problem the night before it is due is a recipe for failure.

Policies to keep in mind

- **You may use any source at your disposal**—paper, electronic, or human—but you *must* cite *every* source that you use, and you *must* write everything yourself in your own words. See the academic integrity policies on the course web site for more details.
- **Being able to clearly and concisely explain your solution is a part of the grade you will receive.** Before submitting a solution ask yourself, if you were reading the solution without having seen it before, would you be able to understand it within two minutes? If not, you need to edit. Images and flow-charts are very useful for concisely explain difficult concepts.

See the course web site (<https://ecealgo.com/su26>) for more information.

If you have any questions about these policies,
please don't hesitate to ask in class, in office hours, or on Piazza.

1. Give a context-free grammar (CFG) for each of the following languages. You must provide explanation for how your grammar works, by describing in English what is generated by each non-terminal. (Formal proofs of correctness are not required.)

(a) $\{x(110)^n x^R : x \in \{0, 1\}^*, n \geq 1\}$

(b) $\{1^i 0^j 1^k : j = 2i + 3k, i, j, k \geq 0\}$

(c) $\{1^i 0^j 1^k : i + k \text{ is divisible by } 3 \text{ and } 0 \leq j \leq k\}$

2. For each of the following languages, decide whether it is regular, not-regular, context-free, not-context free or some combination. Whatever you choose, prove it (if you say it is regular, show the DFA/NFA/RegEx, if you determine it is context-free show the CFG/PDA, if you determine it is not regular show the fooling set, etc.)!

- (a) Let $\Sigma = \{0, 1, 2\}$ and define the language:

$$L_1 = \{0^n w 1^n : w \in \{2\}^*, n \geq 0\}$$

- (b) Let $\Sigma = \{0, 1\}$ and define the language:

$$L_2 = \{x0y : x, y \in 1^*, \text{ and } |x| > 0, |y| > 2\}$$